

# NABL Document Intelligence

Reading a laboratory's accreditation paperwork and filling the NABL application form it belongs to — without ever writing a value we cannot point back to a page.

# One application, a dozen unrelated documents

A single NABL application draws on paperwork from a dozen different issuers, in a dozen different layouts — many of them scans of paper.

- ◆ GST, CIN and legal registration proofs
- ◆ Equipment calibration certificates
- ◆ Reference-material certificates
- ◆ Staff CVs and authorised-signatory declarations
- ◆ PT/ILC reports and the quality manual

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**15–30**

PAGES IN A FILLED APPLICATION  
FORM

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**9**

REAL FORM VARIANTS, IN TWO  
DIFFERENT SHAPES

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**7.4%**

OF PAGES HAVE NO TEXT LAYER –  
MEASURED

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**17**

DISTINCT SUPPORTING DOCUMENT  
TYPES

# Evidence or nothing

A value that cannot be  
matched back to text  
on a real page never  
reaches the form.

That single rule shapes everything else. The output here is a compliance artifact — “the model said so” is not a defensible provenance in front of an assessor.

## **Preserve, never discard**

Merging records a merge. Conflict resolution marks a preference and keeps every candidate. Rejected values are flagged, not deleted.

## **Every page accounted for**

Blank and unreadable pages get a record too, so an omitted page cannot hide as a page that never existed.

## **Report honestly**

A failed chunk appears in the result. An exhausted budget records why it stopped.

# Eighteen steps in seven phases

|                        |                                                                                                               |
|------------------------|---------------------------------------------------------------------------------------------------------------|
| Document ingestion     | 1 Upload — PDF, image, DOCX 2 Validate the file 3 Store, size cap during write                                |
| Document processing    | 4 Orchestrator starts 5 Page text, OCR where needed 6 Analyse sections & tables 7 Chunk with audited coverage |
| Information extraction | 8 Discover target schema, map by meaning 9 Extract each chunk via Nova — JSON + evidence                      |
| Completeness check     | 10 Page, chunk & evidence coverage 11 Missing? → re-analyse only those chunks · max 2 rounds                  |
| Conflict resolution    | 12 Detect duplicates & contradictions 13 Conflict? → adjudicate, keep every candidate                         |
| Validation             | 14 Schema, rules, evidence against real page text 15 Valid? → correct & reprocess · max 3 attempts            |
| Output & storage       | 16 Final JSON + form 17 Persist across 6 tables 18 Return to frontend                                         |

Checkpointed after every node — a crash at chunk 41 of 80 resumes at 41.

# Eleven of sixteen nodes never call a model

5

NODES THAT USE AN LLM

11

NODES THAT ARE PLAIN PYTHON

Anything decidable by counting pages, comparing strings or checking a schema is decided in code.

◆ **Correctness is testable**

— 83 tests, no network call

◆ **Cost scales with the document**

, not the pipeline

◆ **“Model problem or logic problem?”**

has an answer

**The integrity check is never delegated to the thing being checked.**

Evidence validation, completeness and quality control are all deterministic.

# Three options, two rejected

## A — Single-shot REJECTED

Whole document, text plus every page image, one call.

- Best results on *short* documents
- Hits 25 MB at ~65 pages
- 269 test rows truncated the JSON — nothing parsed
- Cannot say what it read

## B — Schema + retrieval REJECTED

One call per schema section, retrieving only relevant pages.

- Cheapest design that scales
- Only finds what the schema already names
- Retrieval can silently miss a page
- Not resumable

## C — Agentic workflow CHOSEN

16 nodes, typed state, checkpointed after each.

- Scales to any length — no truncation
- Resumable, partial failure survivable
- Recovery *changes strategy*, not just retries
- The audit trail is the product

# A 10-page document costs 2.5 cents

5

LLM CALLS

\$0.025

TOTAL — ABOUT ₹2.23

| WHERE THE 5 CALLS GO       | CALLS |
|----------------------------|-------|
| Extraction — one per chunk | 1     |
| Document analysis          | 1     |
| Name equivalence / merge   | 1     |
| Conflict adjudication      | 1     |
| Form mapping               | 1     |

### Four of the five calls are fixed

Ten pages fit in a single 24,000-character chunk, so orchestration — not extraction — is most of the bill on a small document. This is why cost per page falls sharply as documents grow.

| TOKEN SPLIT       | TOKENS | COST     | SHARE |
|-------------------|--------|----------|-------|
| Input @ \$0.30/M  | 22,005 | \$0.0066 | 26%   |
| Output @ \$2.50/M | 7,500  | \$0.0188 | 74%   |

Output is 25% of the tokens and 74% of the cost. Cost control lives in how verbose the JSON is — not in how much we feed the model.

# Cost per page falls 4× as documents grow

| PAGES | CHUNKS | CALLS | COST     | PER PAGE  |
|-------|--------|-------|----------|-----------|
| 10    | 1      | 5     | \$0.0254 | \$0.00254 |
| 20    | 2      | 6     | \$0.0309 | \$0.00155 |
| 50    | 5      | 9     | \$0.0476 | \$0.00095 |
| 100   | 9      | 13    | \$0.0714 | \$0.00071 |
| 200   | 18     | 22    | \$0.1229 | \$0.00061 |
| 300   | 27     | 31    | \$0.1745 | \$0.00058 |

## Marginal cost of 10 more pages

Inside a document already running: one extra chunk, one extra call, \$0.0056.

## 1,000 ten-page documents

\$25.35 per month. Lambda compute and S3 add roughly 6% on top; the model is 94% of the bill.

## Worst case, within the guards

Both re-analysis rounds consumed adds 50%. The three loop guards are what make that a ceiling rather than a hope.



# Well-built, and unproven

83

GRAPH TESTS, ALL OFFLINE

285

TESTS ACROSS THE REPOSITORY

0

PRODUCTION ACCURACY FIGURES

Gemini

WHAT IT ACTUALLY RUNS ON TODAY

**Nova is entitlement-blocked** **BLOCKING**

Bedrock returns `ValidationException: Operation not allowed` for every model on the account. Not a code defect — but it means no benchmark has run.

**Bounding boxes were lost** **REGRESSION**

The graph works from page text, so the review UI shows a page-level highlight instead of a tight box.

**Extraction is text-only**

We gave up a page-image cross-check that measurably suppressed a hallucination class. Worth revisiting at 230 tokens per page.

# Three things, in this order

## 1 Get a number

Unblock the Bedrock entitlement and run the labelled corpus through the scoring harness. Both already exist. Until this lands, every accuracy claim is unmade.

## 2 Put the image back

Send the page raster alongside text for chunks whose OCR confidence is poor. 230 tokens per page against a hallucination class we know it fixes.

## 3 Recover the boxes

Thread PyMuPDF span geometry through chunking and match evidence back to spans, restoring the tight highlight the review UI was designed around.

**The question for this room:** is a page-level highlight good enough to put in front of a reviewer while item 3 is outstanding — or does that block a pilot?

# Numbers worth remembering

|                                                         |                                                         |                                                 |                                                    |
|---------------------------------------------------------|---------------------------------------------------------|-------------------------------------------------|----------------------------------------------------|
| <div>5</div> <div>CALLS PER 10-PAGE DOCUMENT</div>      | <div>\$0.025</div> <div>COST PER 10-PAGE DOCUMENT</div> | <div>22</div> <div>CALLS AT 200 PAGES</div>     | <div>\$0.12</div> <div>COST AT 200 PAGES</div>     |
| <div>16</div> <div>WORKFLOW NODES — 5 USE A MODEL</div> | <div>24,000</div> <div>CHARACTERS PER CHUNK</div>       | <div>3</div> <div>INDEPENDENT LOOP GUARDS</div> | <div>2,000</div> <div>MAX PAGES PER DOCUMENT</div> |
| <div>7.4%</div> <div>PAGES NEEDING OCR — MEASURED</div> | <div>74%</div> <div>OF COST IS OUTPUT TOKENS</div>      | <div>9</div> <div>NABL FORMS SUPPORTED</div>    | <div>0</div> <div>ACCURACY FIGURES — YET</div>     |

# What would you want proven first?

Full written report, cost model and appendices are in `docs/NABL_Project_Report_Rev2.pdf`.